

SQL PROJECT

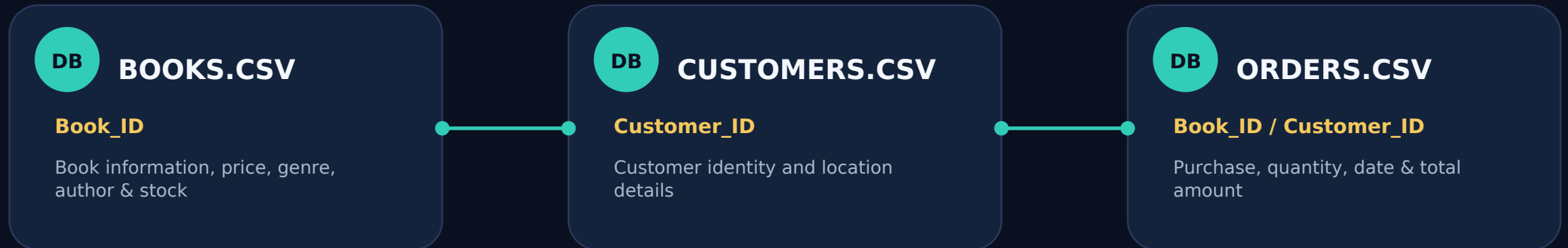
ONLINE BOOK STORE

Databasedesign • SQLqueries • Business analysis

20 practical questions with ready-to-run SQL



DATASET & TABLE RELATIONSHIPS



JOINING RULE

Tables must share a common column with the same name and data type.

BASIC QUERIES

Core retrieval, filtering, sorting and aggregation tasks

- 01 Retrieve all books in the "Fiction" genre
- 02 Find books published after the year 1950
- 03 List all customers from the Canada
- 04 Show orders placed in November 2023
- 05 Retrieve the total stock of books available
- 06 Find the details of the most expensive book
- 07 Show all customers who ordered more than 1 quantity of a book

ADVANCED QUERIES

Aggregations, joins, grouping, ranking and inventory analysis

- 12** Retrieve the total number of books sold for each genre
- 13** Find the average price of books in the "Fantasy" genre
- 14** List customers who have placed at least 2 orders
- 15** Find the most frequently ordered book
- 16** Show the top 3 most expensive books of 'Fantasy' Genre
- 17** Retrieve the total quantity of books sold by each author
- 18** List the cities where customers who spent over \$30 are located

QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q1

Retrieve all books in the "Fiction" genre

```
SELECT * FROM Books  
WHERE Genre='Fiction';
```

Q2

Find books published after the year 1950

```
SELECT * FROM Books  
WHERE published_year > 1950;
```

Q3

List all customers from the Canada

```
SELECT * FROM Customers WHERE  
country = 'Canada';
```

QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q4

Show orders placed in November 2023

```
SELECT * FROM Orders  
WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';
```

Q5

Retrieve the total stock of books available

```
SELECT SUM(stock) AS Total_Stock FROM Books;
```

Q6

Find the details of the most expensive book

```
SELECT * FROM Books ORDER BY  
Price DESC LIMIT 1;
```

QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q7

Show all customers who ordered more than 1 quantity of a book



```
SELECT * FROM Orders WHERE  
quantity>1;
```

Q8

Retrieve all orders where the total amount exceeds \$20



```
SELECT * FROM Orders WHERE  
total_amount>20;
```

Q9

List all genres available in the Books table



```
SELECT DISTINCT genre FROM Books;
```

QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q10

Find the book with the lowest stock

```
SELECT * FROM Books  
ORDER BY stock LIMIT 1;
```

Q11

Calculate the total revenue generated from all orders

```
SELECT SUM(total_amount)  
AS Total_Revenue from Orders;
```

Q12

Retrieve the total number of books sold for each genre

```
SELECT b.Genre, SUM(o.Quantity) AS Total_Books_sold FROM Orders o  
JOIN Books b ON o.book_id = b.book_id
```


QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q13

Find the average price of books in the "Fantasy" genre

```
SELECT AVG(price) AS Average_Price FROM Books  
where genre = 'Fantasy'
```

Q14

List customers who have placed at least 2 orders

```
SELECT o.customer_id, c.name, COUNT(o.Order_id) AS ORDER_COUNT FROM orders o  
JOIN customers c ON o.customer_id=c.customer_id GROUP BY o.customer_id, c.name  
HAVING COUNT(Order_id) >=2;
```

Q15

Find the most frequently ordered book

```
SELECT o.Book_id, b.title, COUNT(o.order_id) AS ORDER_COUNT FROM orders o  
JOIN books b ON o.book_id=b.book_id
```

QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q16

Show the top 3 most expensive books of 'Fantasy' Genre

```
SELECT * FROM books
WHERE genre ='Fantasy'
ORDER BY price DESC LIMIT 3;
```

Q17

Retrieve the total quantity of books sold by each author

```
SELECT b.author, SUM(o.quantity) AS Total_Books_Sold FROM orders o
JOIN books b ON o.book_id=b.book_id
GROUP BY b.Author;
```

Q18

List the cities where customers who spent over \$30 are located

```
SELECT DISTINCT c.city, total_amount
FROM orders o
JOIN customers c ON o.customer_id=c.customer_id
```

QUESTIONS & SQL QUERIES

Enlarged question text + clearer SQL code

Q19 Find the customer who spent the most on orders

```
SELECT c.customer_id, c.name, SUM(o.total_amount) AS Total_Spent FROM orders o
JOIN customers c ON o.customer_id=c.customer_id GROUP BY c.customer_id, c.name
ORDER BY Total_spent Desc LIMIT 1;
```

Q20 Calculate the stock remaining after fulfilling all orders

```
SELECT b.book_id, b.title, b.stock, COALESCE(SUM(o.quantity),0) AS Order_quantity,
b.stock- COALESCE(SUM(o.quantity),0)
AS Remaining_Quantity FROM books b LEFT JOIN orders o ON b.book_id=o.book_id
GROUP BY b.book_id ORDER BY b.book_id;
```

THANK YOU

QUESTIONS • QUERIES • DATABASE ANALYSIS

